



Smart system for water quality monitoring utilizing long-range-based Internet of Things

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Abstract

Water is the most basic need for humans and a source of livelihood for humans. Lack of human awareness to maintain water quality, causing water to become polluted, by both industrial and household waste, impacts on human health and material loss. Thus, it is important to create technology that can monitor water pollution automatically and quickly. This research aims to create a system which utilizes the Internet of Things (IoT) technology that can facilitate quality of water by measuring parameters such as pH and turbidity. The methodology of the system progresses by the usage of a controller which is ATmega328P-AU, pH sensor to measure acidity, turbidity sensor to measure turbidity level, LPWAN LoRa works like a communication of data transmission as well as cloud service, namely Antares, to store data that are sent via Android. Based on the outcomes, the proposed system has achieved a reliable accuracy with percentage error of 99.73% in pH sensor and 99.41% in the turbidity sensor. Also, 2.6 s is the average required time to deliver the results to the cloud service.

Keywords Water quality · Internet of Things (IoT) · Android · LPWAN LoRa · Cloud service

Introduction

One of the most important natural sources is water which is gifted to the human (Najah Ahmed et al. 2019). However, the water resources have been deteriorated and contaminated in increasing pace due to the advancements of community and considerable amount of people tasks. Earth's surface is covered by 71% of water, whereas freshwater is just 2.5%

where lakes and rivers contain only 0.3% (Musambachime 2005). Indeed, freshwater is finite source which influence surrounding environments involved in lakes, rivers and the goodness of food that is consumed (Yahya et al. 2019).

Throughout human activities, for example waste or industrial pollution, the rivers' and lakes' water could be dangerously contaminated (Madadi et al. 2015; Najah et al. 2011). Such activities could lead to death, or at least illness,

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of animals which consume, or live in, water (Najah et al. 2011). Moreover, this would influence of vegetation growth surround that specific area where it could additionally lead to harm the people throughout disease (Ighalo et al. 2021).

In order to provide safety to our surrounding environment, it is highly required to monitor pollution level of water, air, etc. It is needed to purify the polluted elements once it surpasses the safety limits. As known, “Prevention is better than cure,” consequently it is needed to be alerted prior the pollution level surpasses the safety lines (Saha et al. 2017). For water pollution, it can occur anywhere and is rapidly expanding (Bird et al. 2019). The causes of water pollution are also numerous, including household waste, factory waste that uses chemicals, as well as waste from rubbish that is dumped carelessly into rivers. Based on reports from the Director-General of Environmental Control and Damage of the Ministry of Environment and Forestry (KLHK), as much as 68% water quality of river in 34 Indonesian provinces was greatly polluted throughout 2015 (Bank 2016).

Indeed, at all times consumable water is the most needed element for all creatures, including human; thus, it is highly needed to overcome all the obstacles that it faces in real-time operation. Reasons behind these obstacles revolve around growing population, aging infrastructure, etc. Thus, it is necessary to proceed forward more advance approaches for water quality monitoring (Vijayakumar and Ramya 2015). Furthermore, for better quality of water monitoring, it is essential to identify every modification in the pattern of water quality variables from period to period in order to ensure its purity in real time (Das and Jain 2017).

Quality of water is required to follow certain laws and terms which are needed to be consistently monitored (Chen et al. 2020). Hence, Internet of Things (IoT) enables modernistic approaches as well as probabilities to maintain the water quality (Alahi et al. 2018). Cheap gadgets which are linked together capable to be functionalized in particular areas like rivers, lakes and close to the coastline or nearby the industry complexes (Khatri et al. 2018). IoT capable regularly keep track of water variables and by integrating all linked devices, it will create a system that will play an important role in alerting the authorities immediately once abnormal changes occur. IoT in the form of new intelligent system which is able to observe the quality of water in real time as well as provide clear interpretation of the data (Dupont et al. 2018). This approach is reliable, suitable and inexpensive technique to provide management of environment and mitigate human health cases (Maher 2018). Indeed, nowadays, almost all of approaches utilized in the favor of quality of water assurance include physical sampling and then via costly laboratory tests. The replacement is to automatically progress via cheap sensors implementation

located in points and progress of the data which is collected. Therefore, a quality monitoring system of river water was created through adapting the IoT technology that can monitor quickly and efficiently.

Manoharan and Rathinasabapathy (2018) have carried out an experimental research that has utilized Internet of Things and wireless sensor network for monitoring quality of water. This suggested approach possesses low power, low cost, scalable and long range for monitoring of quality of water using LoRa module. Cloete et al. (2016) proposed a monitoring system of water quality via utilizing of carb farming via IoT approach toward providing information to farmers to control reliable quality of water levels in the pond. This approach has contributed toward boosting the rate of crab survival and obtain greater soft shell crab yield (Niswar et al. 2018). Irawan et al. (2021) suggested a monitoring system for water quality for ponds of aquaculture that depends on the low-range internet of things (NB-IoT) approach. Yue et al. (1997) has used IoT for the real-time water quality monitoring integration which has provided a friendly user interface besides infographics as well as a simple approach to the communication with sensors and to configure the controllers. The Citarum River in the Bandung Basin of Java Island, Indonesia, is well-known around the world for its appalling pollution. The official Indonesian water quality control guidelines are exceeded by a variety of pollutants. Devices for water quality measurements can be expensive, particularly for those living in rural and remote communities. In addition to that these devices require regular maintenance and repair, which can be difficult in areas with limited access to trained personnel and spare parts. Therefore, there are needs to advance technique such as IoT monitoring systems which can be extremely beneficial to rural communities in terms of monitoring water quality. By providing real-time data on water quality, rural communities can take proactive steps to protect their water source and ensure it remains safe for drinking and other uses. Finally, by providing real-time data, rural communities can make better informed decisions about water management and conservation (Hartono et al. 2022; Manoharan and Rathinasabapathy 2018). Two parameters widely investigated by researchers, pH and turbidity of being more easily and affordably analyzed than microbiological parameters and can also be assessed using online equipment (Cloete et al. 2016). A study carried out by Irawan et al. (2021) developed monitoring system for measuring three parameters including pH and turbidity. Hence, for quality monitoring system to this river, it needs a sensor of water pH as well as a sensor of turbidity that are connected toward the microcontroller as the controller and is connected to a cloud service namely (Antares) to store data for displaying purpose on Android. Therefore, this study aims to create, by utilizing IoT, a device that is capable of measuring

water quality parameters in the form of pH and turbidity on a regular basis to monitor water quality. Furthermore, the proposed device is created to be progressed continuously and quickly. The methodology of the proposed device is that LoRa is used to send all measurement and processing using an Arduino Uno microcontroller which is combined in a printed circuit board (PCB). All measurement data are sent to Antares as a cloud service to store data and then display it on an Android-based smartphone. Then, the results of the proposed system will be compared with actual measurements in order to examine the reliability of the accuracy level that the proposed system has achieved.

Internet of Things

Internet of Things is a computational notion which describe wide-ranging linkage to the Internet, converting normal appliance to linked gadgets (Yue et al. 1997). The main cue lies beyond Internet of Things' concept is hiring enormous number of smart devices that are able to sense the surrounding environment, transmit and process data that is collected, followed by comment on the environment (Sisinni et al. 2018).

As illustrated by Gunturi et al. (2018) in Fig. 1, the IoT is a conceptualization in which specific items possess the capability to convey data to a network with no need for person-to-person interaction or from person-to-computer gadgets. The IoT has developed quickly from the wireless convergence technology, the Internet as well as micro-electromechanical systems (MEMS). According to Asthon (2010), IoT was first

introduced by Suresh et al. (2018). The concept of IoT can be interpreted as an ability to expand the benefits of continuously connected internet connectivity. IoT has capabilities like sharing data, remote control, etc. All are connected to the network through embedded sensors and are always active (Suresh et al. 2018).

Low-power wide area network LPWAN is representing a novelty of communication model, which would supplement old cellular as well as short-range wireless approaches in delivering variety of IoT applications' needed specifications (Hernandez et al. 2017). LPWA technologies provide unparalleled groups of characteristics involving wide-range area connection in low data and low power rate devices (Raza et al. 2017). It is a sort of wireless telecommunications network prepared for long distance that works on a machine to machine network coverage, with wider range, requirements of low power, as well as less development expenses of cellular networks, and LPWAN can become an IoT network backbone where each object can be connected and communicate without human intervention (Nurliyana Kafli 2017).

Referring to Fig. 2, LPWAN capable of widely categorized into two categories called licensed LPWAN (such as 5G IoT, EC-GSM-IoT, LTE-M-IoT, as well as NB-IoT), and free licenses or LPWAN which are not licensed (RPMA, LoRa, Telensa, and SigFox) (Augustin et al. 2016).

LoRa (long range) is a modulation technique that allows the transfer of long-distance information using VHF/UHF radio frequencies (Khutsoane et al. 2017). LoRa was patented by Semtech (2019). The application of this LoRa varies from various fields, especially within the area of IoT, and LoRa is used in M2M (machine to machine) communication

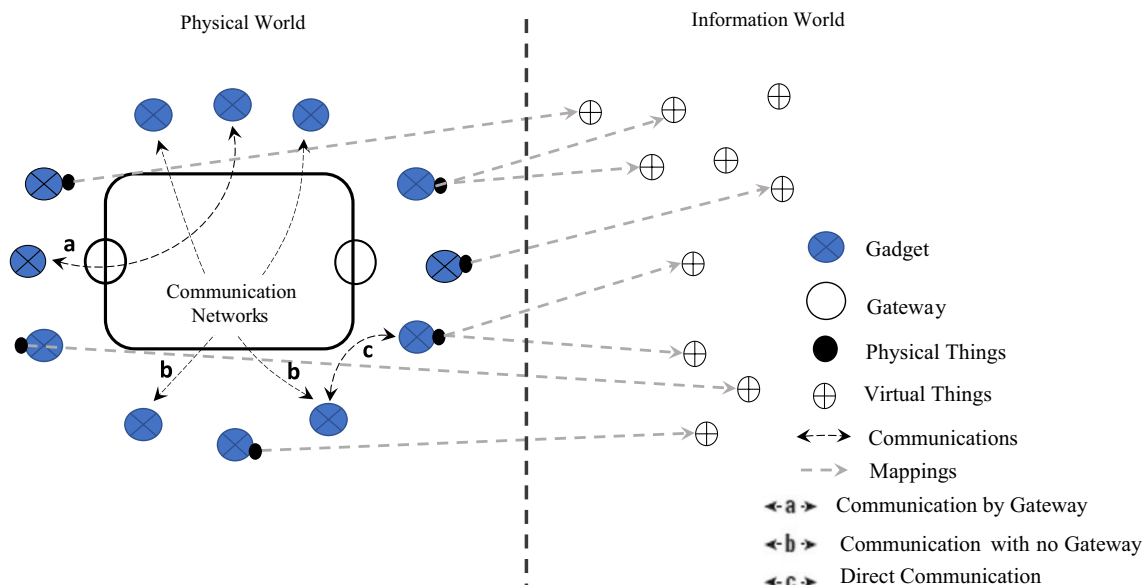


Fig. 1 Concept of IoT

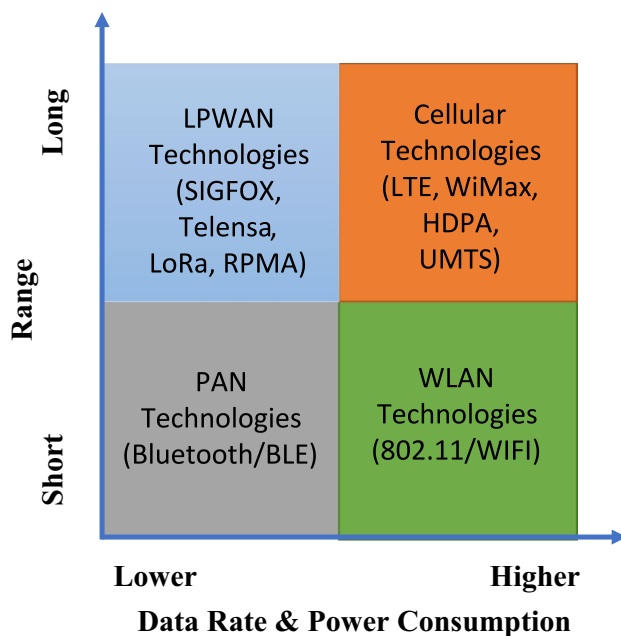


Fig. 2 Range for wireless, power consumption, and data rate communication technology comparison

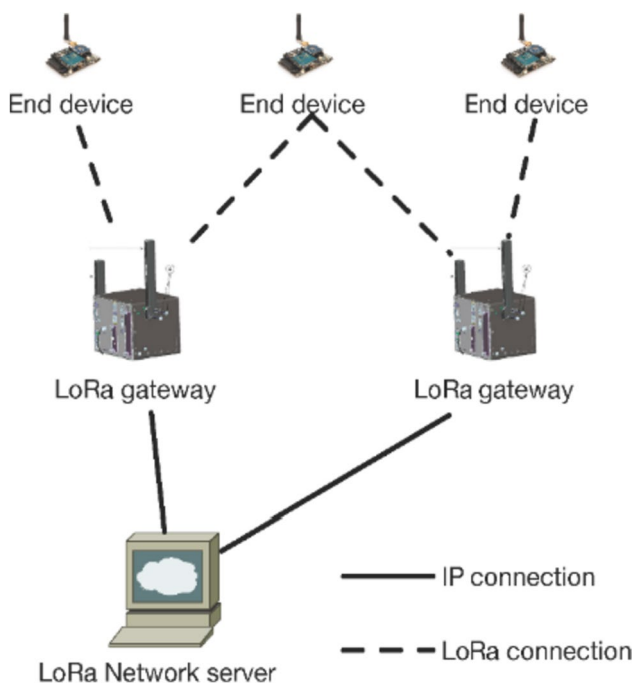


Fig. 3 LoRa architect network

or is intended for use in “Long-lived battery-powered devices” (Kodali 2017).

Referring to Fig. 3, LoRa architect network into three devices:

- Nodes or end devices, which act as data senders.
- Gateway, which acts as a successor to raw data sent from end devices using assistance from networks such as Ethernet, 3G, and 4G LTE.
- LoRa network server is a device where data sent over the LoRa network is stored. LoRa network server can be a personal web server, or a cloud server such as ThinSpeak.

Antares is a brand under PT Telekomunikasi Indonesia that specifically operates on the IoT platform, and antares.id is a web that provides Internet of Things technology services to various applications that serve various types of consumers. The service is a cloud that can store data and can display the results of the parameter data that has been tested.

Design system

The utilization of Internet of Things (IoT) technology is highly needed to determine water pollution automatically and quickly due to water pollution impact on humans' health and loss of material as well as the difficulty of human supervision. This study has developed a monitoring system for water quality which capable of providing water quality in a river utilizing the technology of Internet of Things utilizing communication of LPWAN LoRa. This targets to relieve boost awareness of public regarding the necessity of healthiness water quality. Indeed, water quality is beneficial when it is without the presence of microorganisms, tasteless and colorless (IRLANE MAIA DE OLIVEIRA 2017).

According to Marselina et al. (2022), condition of river water is categorized based on water quality index:

- *Excellent* Quality of water meets every criterion to be utilized as a clean water source.
- *Good* Quality of water narrowly or rarely violates criteria to be used as a clean water source.
- *Fair* Quality of water sometimes violates criteria, perhaps via a huge margin, to be used as a clean water source.
- *Bad* Quality of water usually violates criteria to be used as a clean water source via a significant margin.
- *Very bad* Quality of water does not suit any of clean water source criteria.

In Indonesia, nine factors are taken into consideration by the National Sanitation Foundation Water Quality Index (NSF WQI): turbidity, total solids, temperature, pH, DO, BOD, nitrate, total phosphorus, and fecal coliform (Marselina et al. 2022). In this study, two parameters were chosen (pH and turbidity). In addition to being more easily and affordably analyzed than microbiological parameters, these parameters can also be assessed using online equipment (Cloete et al. 2016).

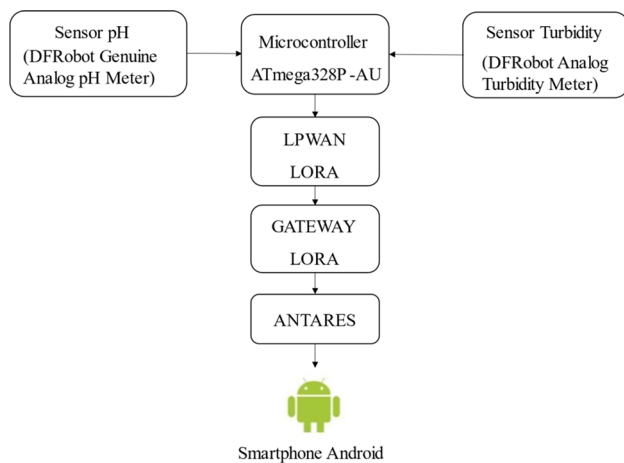


Fig. 4 Diagram of system block

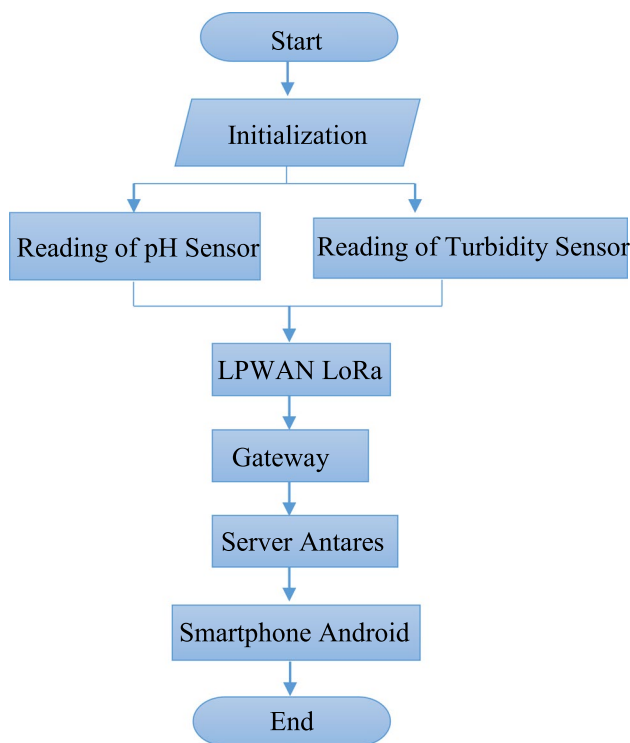


Fig. 5 System flowchart

This study system, which is illustrated in Fig. 4, is developed utilizing a microcontroller called ATmega328P-AU which is linked with various sensors for capturing predetermined variables such as pH and turbidity air. ATmega328P-AU runs the controller; after that by utilizing the module of LPWAN LoRa communications, the measured data pass to the Antares cloud service from the sensor. To be possibly

approved on an Android smartphone, the measured data shall be processed in the cloud database.

As described in Fig. 5, the proposed approach shall initiate, once linked to a source of voltage as well as linked to the initialization of internet connectivity. The sensor shall identify and capture the variable in which the outcomes of the reading data of the sensor shall be progressed utilizing a microcontroller. After that, the size of data shall be passed by LPWAN LoRa toward LoRa gateway that shall be kept within the Antares database. The reading data outcomes shall be kept in the cloud in which could be shown on Android smartphone.

ATmega328P-AU

It consists of an 8-bit CMOS low-power microcontroller relied upon the structure of RISC which empowers AVR via progressing robust tracking in one period. It reaches the outcomes of 1 MIPS per MHz which the proposed technology needs an advancing in the consumption of power against increased speed. ATmega328P-AU an amalgamated circuit, besides its capability of a computer in a programmable chip, can manage peripherals such as LEDs, servos, motors and sensors (Cain and Cross 2018).

LoRa RFM95

The long-range low-power transceiver model RFM95 possesses a spectrum communication spread which is an extremely long range. This is capable to achieve a longer distance from the conventional technique relied on FSK modulation. The RFM95 LoRa model has been widely utilized like the transceiver to receive as well as transmit signals (Dala and Arslan 2021). The model has the highest flexibility for consumers to identify on error correction rate (CR), spread spectrum modulation bandwidth (BW) and spreading factor (SF). RFM95 LoRa specifications are programmable bit speeds of up to 300 kbps, +14 dBm PA high efficiency, high sensitivity –111 to –148 dBm, effective bitrate 18–37.5 kbps, bandwidth 7.8–500 kHz, spread factors 6–12 and 868/915 MHz frequency range.

pH sensor

The pH sensor (DFRobot Analog pH Meter) serves to measure acidity or basicity in the surrounding water. The pH functionality revolves around the alkalinity's or acidity's measurement of a solution. Its scale is ranged from 0 to 14, whereas 7 is a neutral point. Values higher than seven denotes an alkaline or basic solution; meanwhile,

Table 1 Description of pH sensor

Element	Descriptions
Power supply	5.00 VDC
Signal output	Analog
Dimensions	43° 22 mm
Measurement range	0–14 pH
Accuracy	0.1 pH (25C)

values lower than seven denotes an acidic solution (Fadel and Shujaa 2020). As stated in Table 1, the pH sensor specifications are the power supply 5 VDC, signal output analog, size 43° 22 mm, measurement range 0–14 pH, and accuracy 0.1 pH (25C).

Turbidity sensor

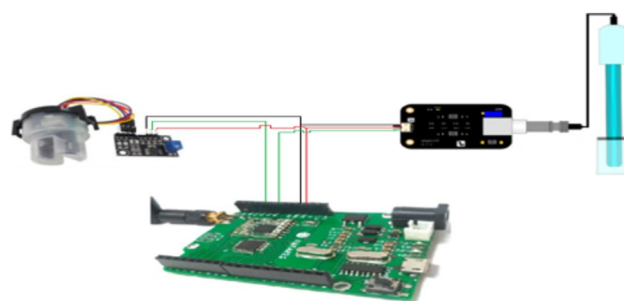
Turbidity is a water cloudiness measurement. Turbidity is used to identify the degree where loses transparency of water occurs. Turbidity is categorized as a good water quality measurement (Parra et al. 2018). The turbidity sensor (DFRobot Analog Turbidity) serves to measure acidity or basicity in the medium of water. As stated in Table 2, the turbidity sensor specifications is power supply 5 VDC, signal output analog, size 38 × 28 mm, measurement range 0–300 NTU, and accuracy 0.1 NTU.

Wiring system

Black box made of acrylic is chosen as the main material for making boxes and packaging of this tool. With the consideration that it is not easy to get wet and water does not quickly get into the black box, and also the price is quite affordable. Board, which is demonstrated in Fig. 6, utilizes a DFRobot analog pH sensor (pH) connected with an analog pin, and DFRobot analog turbidity sensor (Turbidity) is linked to an analog pin. Formerly, pH sensor and turbidity sensor should be calibrated to obtain reliable outputs in pH and turbidity levels.

Table 2 Turbidity sensor's specifications

Element	Descriptions
Power supply	5.00 VDC
Signal output	Analog
Size	38 mm × 28 mm × 10 mm
Measurement range	0–300 NTU

**Fig. 6** Wiring sensor

Software system design

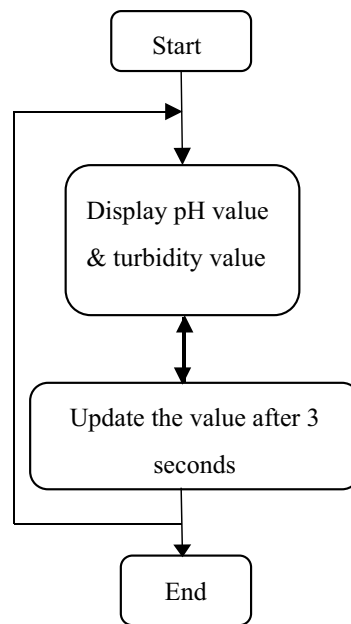
In this system, there are two water diagrams, namely the programming flow diagram on the microcontroller and android application design display from android. This is because the programming on the microcontroller is only limited to the data read via the sensor and followed by passing them toward the cloud, while the programming on the android is useful to get data from the cloud. Figure 7a describes the process of the android application. It describes the flowchart of the program embedded in the ATmega328-AU controller which works from initiation to sending sensor read data to the Antares cloud. The thing that needs to be done is to initialize the access point to connect to the internet and then determine the pin used. Then, the sensor data were uploaded to the cloud. Programming was done using Arduino IDE (Arduino pro or pro mini ATmega328P 3.3 MHz). Figure 7b describes how, after getting the data from the Antares cloud, the output is displayed in Android. Android programming has two programming languages, namely using Java as the main programming language to run the process of the application and XML as programming to create the design of the application display.

LoRa

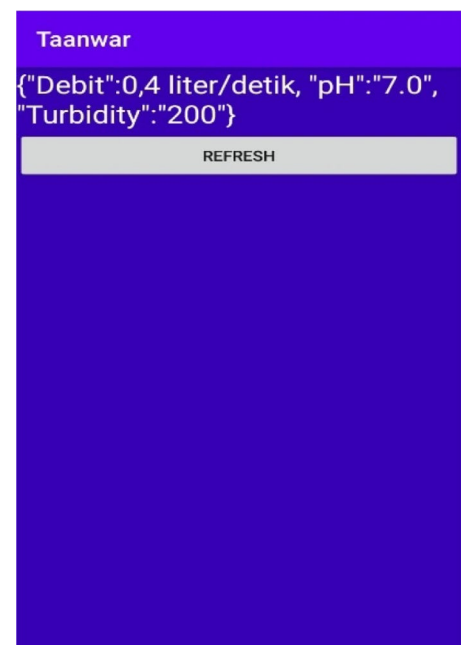
Based on the Regulation of the Minister of Communication and Informatics of the Republic of Indonesia Number 1 of 2019 concerning the use of radio frequency spectrum, it is stipulated that for LPWA LoRa to operate at a frequency of 920–923 MHz, the maximum effective isotropic radiated power (EIRP) for the gateway and end node, respectively, 400mW and 100mW, and a maximum duty cycle of 1%.

The duty cycle will limit the time interval for sending data. The channel occupation time for sending measurement data is about 100ms. When using a duty cycle of 1%, the permitted time interval for sending data is every 10 s. When measuring river water quality, a short sampling period is

Fig. 7 **a** Application flowchart process. **b** Android application design display



(a)



(b)



Fig. 8 River water quality monitoring system

optional. Measurement every 5 min is enough to monitor the condition of the river water in real time. So that each communication channel on LoRa will be able to serve 3000 sensor nodes with reasonable media access control. However, in the case of measuring river water quality, the number of nodes needed to monitor the river as far as the gateway's coverage radius of 5 km requires 50 nodes. So there is no need for special arrangements for access to communication media.

Result and discussion

The results of the quality monitoring system of river water, which are illustrated in Fig. 8, that have been carried out will be tested and analyzed. The test that is carried out serves to get the results of each system that has been designed and realized. The data obtained from the test results are then analyzed in order to determine the extent of the suitability of the system to the target. Testing is implemented to ensure the functionality of the system. Therefore, various tests were conducted.

For pH sensor, testing aims to determine whether the sensor is working according to the expected standards or not. Tests are carried out to equalize the results of the measurement value on the pH meter (Orion pH/ISE meter) and the pH sensor. Throughout the testing process the accuracy of the pH sensor, first measuring the distilled water by the sensor and the pH meter to find out the difference between the sensor measurement and the pH meter in measuring distilled water. Furthermore, ten measurements are made via the sensor and one measurement via the pH meter with the specific fluid. The ten measurements' results via the sensor are then averaged in order to be compared with the pH meter metric results to identify how much the percentage error is generated. Tests have been implemented thirty times. The data acquired from the water pH's outcomes are utilizing DFRobot analog pH meters and Orion pH/ISE meters as illustrated in Fig. 9.

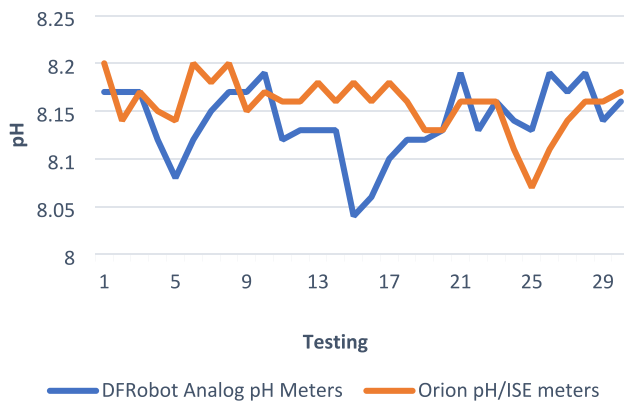


Fig. 9 DFRobot analog pH meters and Orion pH/ISE meters sensor graph

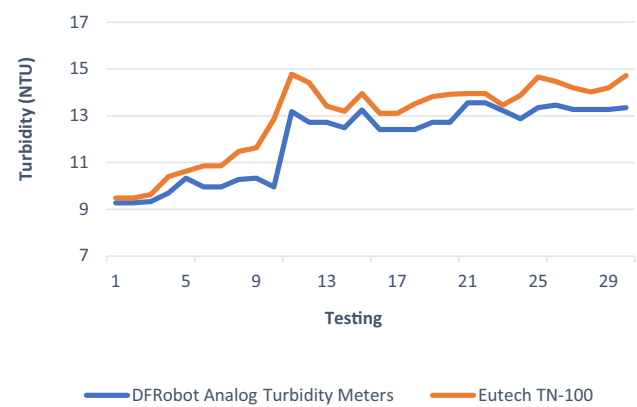


Fig. 11 DFRobot analog turbidity meters and Eutech TN-100 graph

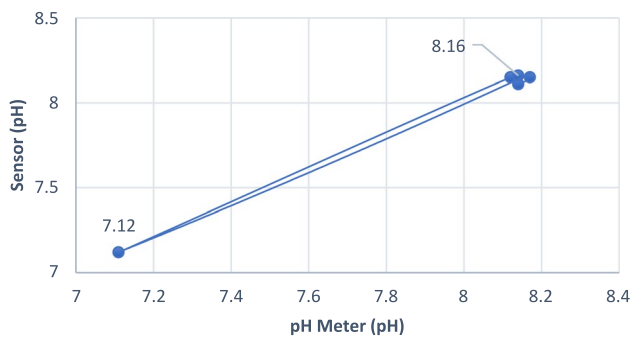


Fig. 10 Turbidity sensor testing graph

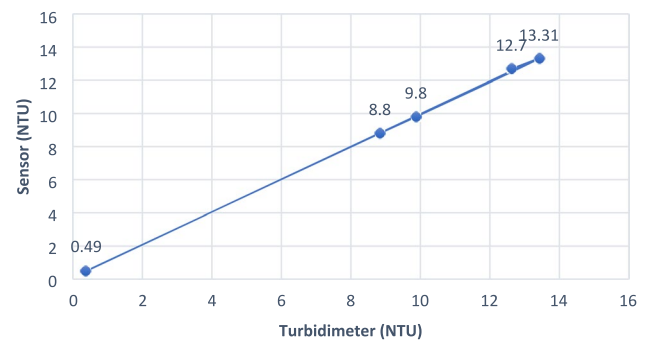


Fig. 12 Turbidity sensor testing graph

The highest values of pH achieved are 8.19, 8.18 and 8.18; meanwhile, the lowest pH values are 8.08, 8.05 and 8.19 in Buah Batu, Sukapura and Sukabirus, respectively. Furthermore, the average value of pH sensor is 8.12, 8.11 and 8.14 in which the percentage error occurred 0.61%, 0.37% and 0.25%. From Fig. 9, it could be shown which water pH measuring the value of the DFRobot Analog pH meter sensor is less than the value of the Orion pH/ISE meter measurement because of the average sensor error value is 0.27%. From the test results above, this sensor can function properly to observe quality of river water in real time, where the results obtained for the pH of water in the Citarum river still meet the requirements for clean water based on the Minister of Health Regulation Number: 416/MEN.KES/PER/IX/1990 because it is still between the maximum and minimum levels allowed (6, 5–9). From the results of sensor testing that has been done, it can be concluded that the sensor has responded to changes in the acidity level (pH) in the measured liquid as shown in Fig. 10. The average percentage of errors obtained is 0.27%. The percentage error is due to sensor sensitivity and ADC conversion. This level of error considers acceptable in comparison with previous study (Irawan et al. 2021).

For turbidity sensor testing, this test is performed to determine the level of accuracy of the sensor. The value generated by the sensor will then be compared with the value from the measuring instrument. The turbidity measuring instrument used is the Eutech TN-100. As for the samples tested are Aquades, river water Citarum in stone fruit, Sukapura, Sukabirus and Babakan Ciamis. In the process of testing the turbidity sensor, ten measurements are made via the sensor and one measurement via the turbidimeter for the self-same fluid. The ten measurements' results via the sensor are then averaged in order to compare them with the measured values. Turbidimeter to find out what percentage of errors are created. The process of testing is implemented on 5 unsimilar fluids. Tests have been implemented thirty times. The data gained from the water turbidity's outcomes utilizing DFRobot Analog Turbidity Meters and Eutech TN-100 as shown in Fig. 11.

As for the results of the sensor accuracy test that has been carried out, it can be concluded that the sensor has responded to changes in the measured fluid turbidity level. When the turbidity in the water increases, the measurement made by the sensor increases as seen in Fig. 11. From Fig. 11, it is clearly shown which water turbidity measuring

that the DFRobot analog turbidity meter sensor value is below the Eutech TN-100 measurement value due to the average sensor error value is 7.02% as illustrated in Fig. 12.

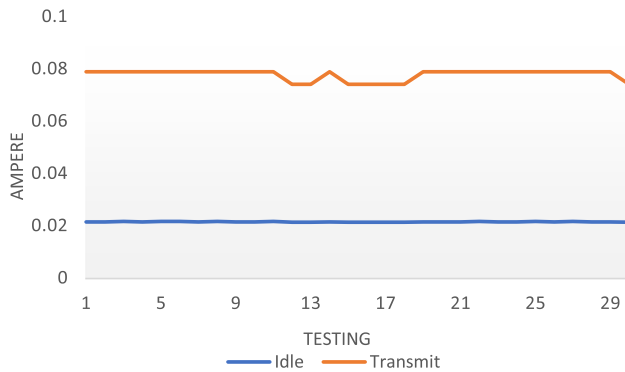


Fig. 13 Idle and transmit power consumption

In order to test of consumption of power targets to identify the overall power required once the gadget is transmitting (sends data) as well as idle (static condition). Size of data relies on the pH sensors measurement results in which tests of turbidity implemented thirty times. The data are achieved from capturing gadgets linked toward power supplies of direct current (DC).

Based on Fig. 13, by using 30 times of testing, the power amount obtained on every condition of transmission of data with a medium of 0.0788 A to transmit and 0.0213 A for idle. Power is a multiplication among voltage and current magnitude. Voltage consumption is utilized in gadgets of 3.3 V; after that, the overall power amount is obtained from every condition: from the table above, we can illustrate it with a graph as a result of the pH sensor response in real time.

To observe quality of river water in real time, as shown in Fig. 14, the results obtained for water turbidity and pH

Fig. 14 Real-time measurement

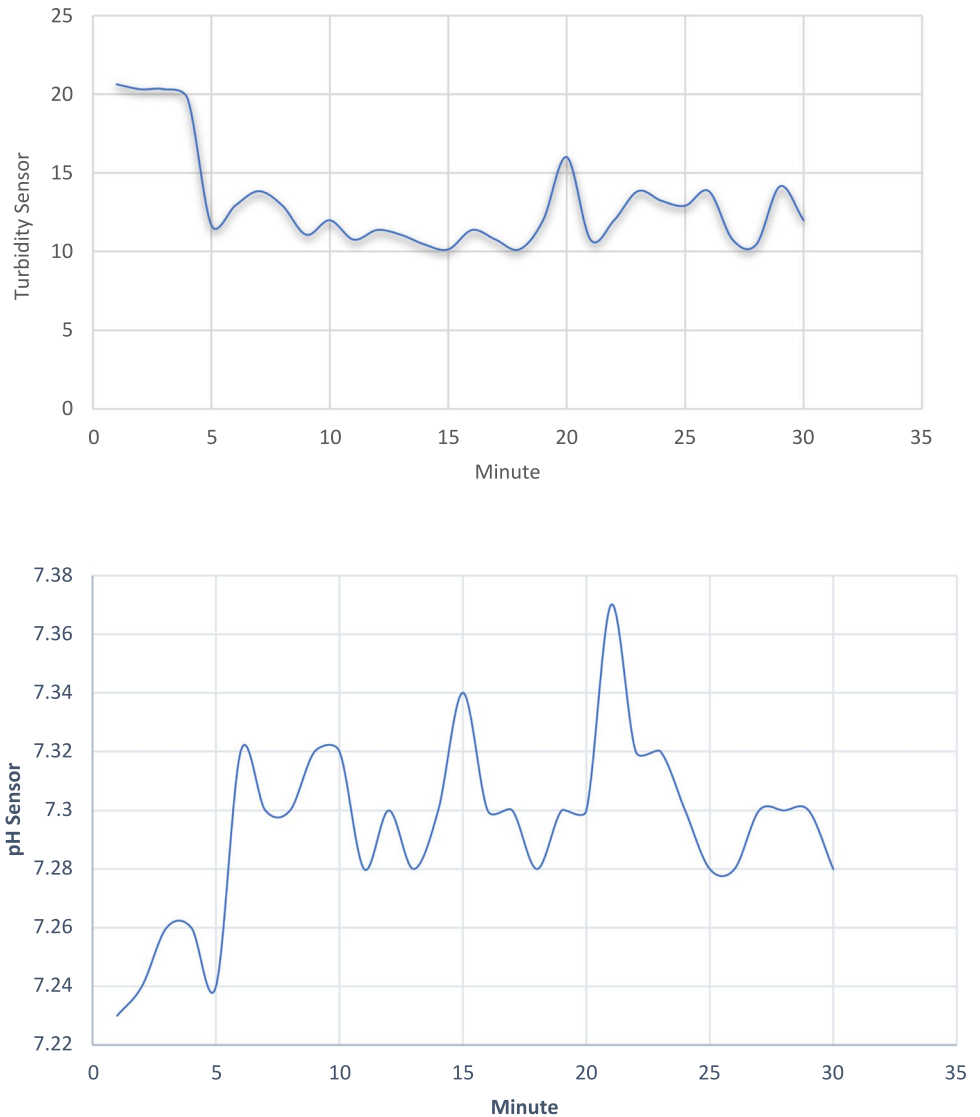
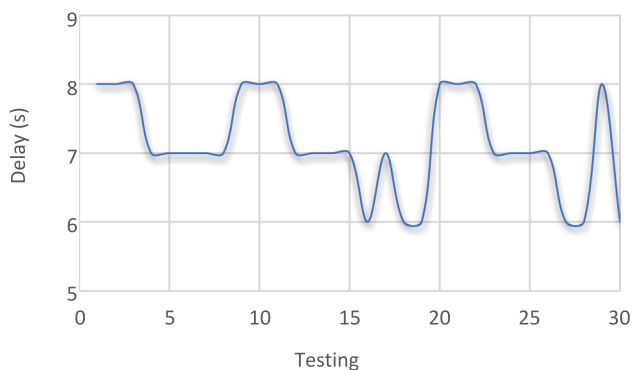


Table 3 Results of power consumption

Conditions	Minimum power	Maximum power
Idle	6996 mW	7095 mW
Transmit	2442 mW	26,004 mW

Table 4 Battery charger

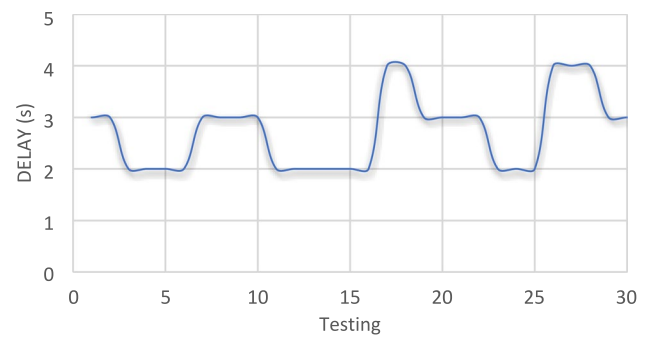
Period	Minimum current	Battery power	Battery
24 h	788 mA	1891 mAh	1/2
168 h	788 mA	13,238 mAh	3
730.001 h	788 mA	57,524 mAh	13
8.760 h	788 mA	690,288 mAh	153

**Fig. 15** Delay time to LoRa gateway

in the Citarum river still meet the requirements for clean water based on the Minister of Health Regulation Number: 416/MEN.KES/PER/IX/1990 because it is still below the maximum allowable value (25 NTU) for turbidity and within the permissible maximum levels that considered between 6.5 and 9 for pH.

Based on Table 3, the results of power consumption from 2 types of conditions are obtained. The average power consumption is in these two conditions; maximum power on idle conditions is 7095 mW and 26,004 mW on transmit conditions. The charger of the battery utilized presuming the capability of a 4500 mAh is stated in Table 4.

With regard to delay and packet loss testing, testing data size, delay, and packet loss to the Antares cloud aims to determine the performance of the system when connected to the internet. Data size testing aims to determine the amount of data capacity measured on the sensor, delay to determine the speed status of the transmitted data, and packet loss to determine the condition of the total number of data packets lost during data transmission. Tests have been implemented thirty times within 5 min. Data gained from the sending

**Fig. 16** Time delay from LoRa gateway to cloud Antares

data's result toward a LoRa gateway where the results are plotted in Fig. 15.

As shown in Fig. 15, the average timing achieved is 7.1 s for the data sending delay toward Cloud Antares. The obtained results showed that the lowest frequency levels achieved is 920,100 MHz with 7 s of delay; meanwhile, the highest frequency is 921,500 MHz with 8 s of delay. Furthermore, the mean result is 920.960 MHz for the frequency and 129.2 bps for bandwidth of the passing data toward the LoRa gateway. Referring to Fig. 16, the mean outcome is 2.6 s for the delayed period which is computed for the data that are sent toward cloud service of Antares from LoRa gateway. Furthermore, it happens single time packet loss on testing to 7; this means that 7 data did not pass to the Antares; however, it is kept in the LoRa gateway. Node's move will not affect power consumption because the transmission power is fixed according to regulations, while the delay in sending data from the node to the LoRa gateway is not much different for distances of 100 m to 5000 m, with results that can still be acceptable for water quality measurements.

Therefore, it can be concluded through comparison of various technologies that LoRa is the most suitable for communication between the sensor nodes and the gateway (Simitha and Subodh Raj 2019). Hence, such system can be adopted as reliable tools for rural communities in monitoring the quality of water.

Conclusion

Due to the water pollution impact on humans' health and loss of material as well as the difficulty of human supervision, the utilization of Internet of Things (IoT) technology is needed to determine water pollution automatically and quickly. Therefore, this research has proposed, with help of IoT technology, a tool which is capable of monitoring the water quality quickly and efficiently through the sensors of pH as well as sensors of turbidity. The sensors are linked to

a microcontroller as a controller and connected to a cloud service, called Antares, to store data for displaying purpose on android. The outcomes of this research are summarized as follows:

1. The designed device has a fairly large level of accuracy of each sensor, namely the pH sensor of 99.73% and the turbidity sensor of 92.98%.
2. The overall consumption of power needed when transmission of data is 244.2 mW for minimum power and 260.04 mW for the highest power.
3. The sending data delay to the Antares cloud is comparatively short since it has an average timing period of 2.6 s. This is possible due to the adjustment of the system from Antares in the data processing process.
4. Referring to the outcomes of water quality testing parameters of Sungai Citarum, an average pH of 8.14 and turbidity of 11.15, so that Sungai Citarum still meets the requirements of clean water quality according to Regulation of the Minister of Health No: 416MEN. KES/PER/IX/1990, namely pH ranged from 6.5 to 9 and maximum turbidity 25 NTU.

As the results of the proposed system are promising, it is highly recommended to utilize the proposed approach in further experimental research. Further work should examine the capability of this system, after necessary development, as flood mitigation tool by the utilization of the water discharge parameter.

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Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval Not applicable

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